

WAZOBIA.NG

A Smart Glove That Speaks Nigeria

TECHNICAL DOCUMENTATION

Architecture, setup and usage of the smart glove and its N-ATLAS integration

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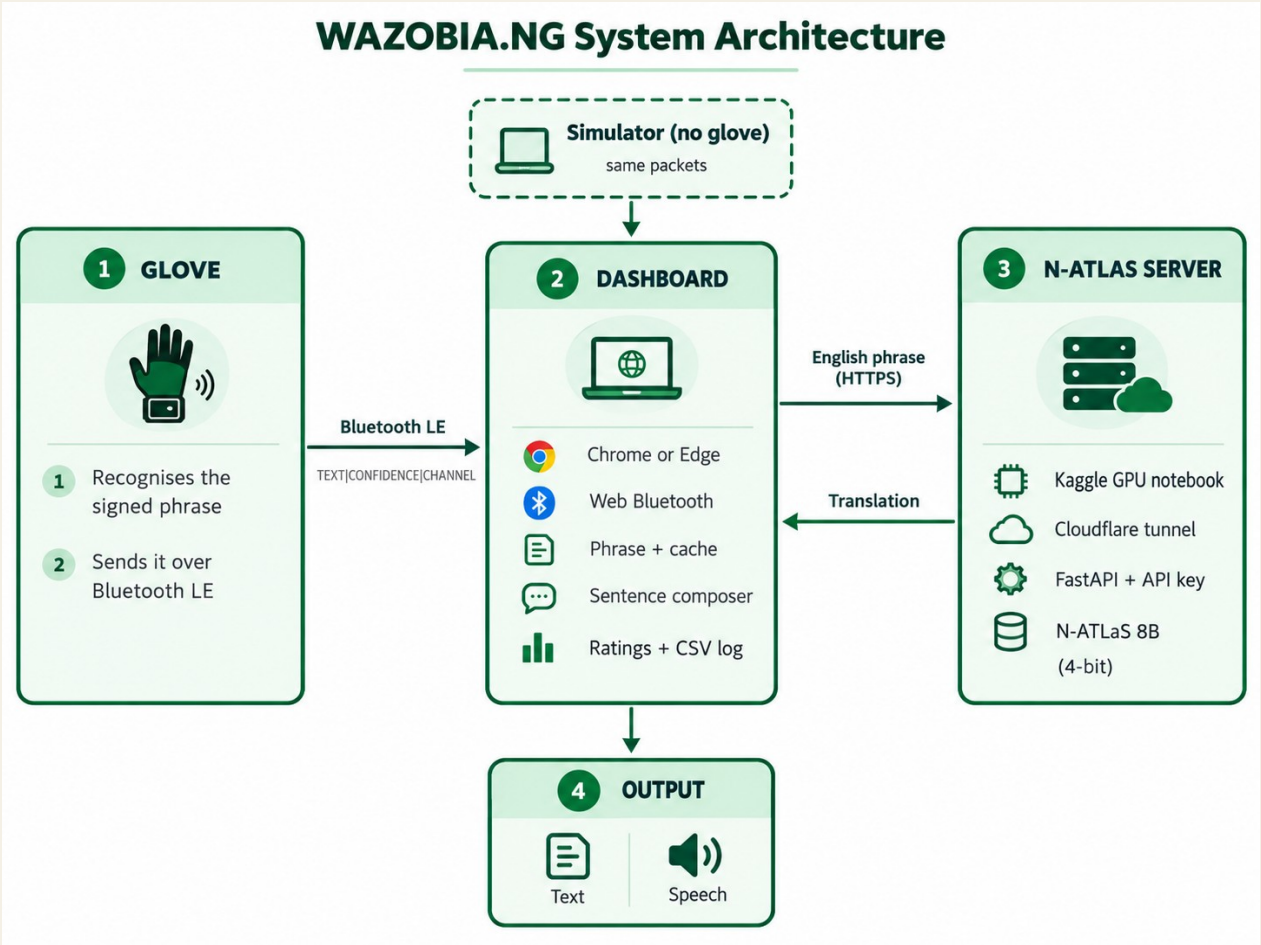


Figure 1. WAZOBIA.NG system architecture.

10	3	8B	4-BIT
PHRASES	NIGERIAN LANGUAGES	N-ATLAS MODEL	GPU DEPLOYMENT

I

Overview

WAZOBIA.NG is a smart glove that recognises selected Nigerian Sign Language phrases. The glove sends each recognised phrase over Bluetooth to a web dashboard, which uses N-ATLAS to translate it into Yorùbá, Hausa or Igbo and speaks the result.

Item	Detail
Vocabulary	10 phrases (two of them are names, which are never translated)
Languages	English, Yorùbá, Hausa and Igbo
Translation	NCAIR1/N-ATLaS (8B), run in 4-bit on a GPU notebook and called by the dashboard
Validation	10 native signers, 120 judged gestures: 95.0% sign accuracy, 84.4% translation accuracy without name gestures (see the Real-World Validation report)

II

System Architecture

The system has four functional parts. The simulator is an optional test path and feeds the dashboard the same packets as the physical glove.

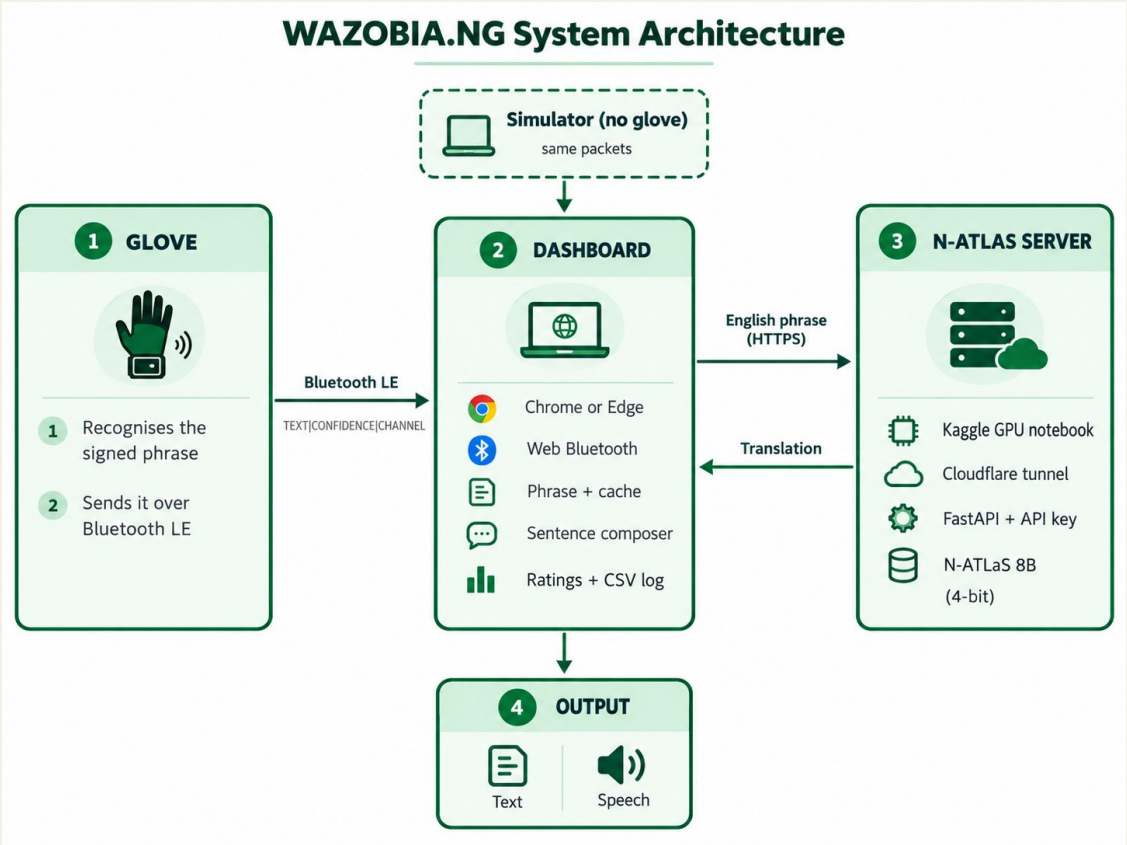


Figure 2. System architecture. The dashed simulator box provides the same dashboard packets as the glove.

- I. Glove. The glove recognises the signed phrase and sends it to the dashboard over Bluetooth LE, together with a confidence value and a gesture number.
- II. Dashboard. The dashboard runs in Chrome or Edge. It receives the packet, requests the N-ATLAS translation, shows and speaks the result, records the interaction and allows a reviewer to rate it.
- III. N-ATLAS server. A GPU notebook loads the N-ATLaS model and exposes a small web API through a temporary public address. The dashboard sends the English phrase and receives the translation.
- IV. Output. The translated text appears on screen and is spoken, using a recorded MP3 when one exists and the browser voice otherwise.

III Glove-to-Dashboard Link

The glove advertises as a Bluetooth LE device whose name starts with WAZOBIA_Glove. The dashboard connects with the Web Bluetooth API in Chrome or Edge and uses the standard Nordic UART Service.

Item	Value
Service	6e400001-b5a3-f393-e0a9-e50e24dcca9e
Glove to dashboard (notify)	6e400003-b5a3-f393-e0a9-e50e24dcca9e
Dashboard to glove (write)	6e400002-b5a3-f393-e0a9-e50e24dcca9e
Gesture packet	TEXT CONFIDENCE CHANNEL, for example Please 97 7 (phrase, confidence in percent, gesture number 1 to 10)
Other messages	The dashboard ignores PONG. It sends MUTE_GLOVE, UNMUTE_GLOVE and VOL:<number> to control the glove audio.

IV N-ATLAS Integration

IV-A. The Server

- I.** A Kaggle notebook (natlas_server_kaggle.ipynb) logs in to Hugging Face and loads NCAIR1/N-ATLAS in 4-bit (NF4) on one GPU, using about 6 GB of GPU memory.
- II.** A FastAPI app offers GET /health and POST /v1/chat/completions. The POST accepts chat messages and returns the reply. It uses a random API key printed by the notebook and handles one request at a time.
- III.** Generation uses temperature 0.1 and a repetition penalty of 1.12, with at most 200 new tokens per request.
- IV.** A Cloudflare quick tunnel provides a temporary public HTTPS address. The address changes every time the notebook is started.

IV-B. What the Dashboard Does

- I.** Prompt. A system message tells the model to reply with only the translation, followed by up to three example phrase pairs from the reference dictionary and then the phrase to translate.
- II.** Names. The two name gestures are shown unchanged and are not sent to the model.
- III.** Cache and warm-up. The first answer for each phrase is kept for the session. When a language is selected, the dashboard translates all ten phrases in the background so later gestures answer instantly.
- IV.** Fallback. If the server does not answer within 25 seconds, the dashboard shows the reference dictionary phrase so the glove keeps working. Each log entry records which source was used.
- V.** Sentence composer. Gestures signed in a row are joined into one sentence. After a 4-second pause, the sentence is sent to N-ATLAS as a whole and the result is shown and spoken.
- VI.** Connection panel. The server address and API key are entered in the dashboard and stored only in the browser. They are never saved in the dashboard file.

IV-C. Why the Model Matters

In the validation sessions, the N-ATLAS wording differed from the stored reference phrase in 59.6% of the judged translations (65 of 109). This shows that the multilingual output is generated by the model rather than copied directly from a lookup table.

V Logging and Ratings

The dashboard records every interaction in the browser and exports it through the Download log button. A reviewer can mark each real-glove gesture as Sign OK and Translation OK. Participants are identified by ID only.

Column	Meaning
id, time, session	Row number, time in UTC, and a random ID for each page load
participant	Participant ID entered on the dashboard
real_use	yes when a participant ID was entered and the input was the glove; otherwise no
input	glove (physical glove) or simulator (Simulator panel)
kind	gesture, sentence (sentence composer) or prewarm (background warm-up)
lang, channel	Language (en, yo, ha, ig) and gesture number (1 to 10)
confidence	Glove confidence in percent (blank for simulated gestures)
english, output	The phrase sent and the text N-ATLAS returned
shown, table_value	The text displayed to the user and the reference dictionary phrase
source	natlas (live call), cache (earlier answer reused), passthrough (name or English) or fallback (server did not answer)
latency_ms	Time taken by the N-ATLAS request (0 when reused)
sign_ok, translation_ok	The reviewer's yes or no ratings

VI Testing Without the Glove

Reviewers who do not have the glove can test the dashboard with the Simulator panel, a collapsed section above the gesture list. It has one button for each of the 10 gestures. Each button sends the same packet the glove sends, so the N-ATLAS call, sentence composer, audio and logging all run normally.

- I.** Simulated gestures are logged with input = simulator and real_use = no, and are excluded from all validation results.
- II.** The simulator does not test sign recognition. That evidence comes from the sessions with native signers described in the Real-World Validation report.

VII Setup and Usage

VII-A. What You Need

- I. A Hugging Face account with access to NCAIR1/N-ATLaS (a gated model) and a read token.
- II. A Kaggle account with a verified phone number so the GPU can be used.
- III. Chrome or Edge on a computer. A powered WAZOBIA glove is required for physical-glove testing, but the simulator can be used without the glove.

VII-B. Start the Server

- I. Create a Kaggle notebook, choose File > Import Notebook and upload code/natlas_server_kaggle.ipynb.
- II. In Settings, set Accelerator to GPU T4 x2 and Internet to On. Under Add-ons > Secrets, add HF_TOKEN with the Hugging Face token and attach it to the notebook.
- III. Run the cells in order. The install cell may print dependency warnings about unrelated packages, which can be ignored. Loading the model takes a few minutes.
- IV. When the tunnel cell finishes, copy the PUBLIC URL and API KEY. The test cell at the end should return HTTP 200 and a translation.

VII-C. Open the Dashboard

- I. Open code/index.html in Chrome or Edge.
- II. In the N-ATLaS connection panel, paste the URL and key and press Save & test. The status should show Server reachable and then Ready once the phrases are cached.
- III. Choose a language (Yorùbá, Hausa or Igbo). Trigger a gesture with the glove using the Bluetooth connection or use the Simulator panel.
- IV. To record a test, enter a participant ID, rate each gesture and press Download log (CSV) at the end.

VII-D. Finish

Stop the Kaggle session when testing is complete so the weekly GPU quota is not consumed while the system is idle.

VIII Limitations and Notes

- I. The N-ATLAS server runs only while a notebook session is open. Its address and key change on every run, so a reviewer starts their own server using the setup procedure above. The demo video and evidence pack show the system working.
- II. Free GPU time is limited. The first test session ran on Colab until the free tier timed out, while later sessions ran on Kaggle.
- III. The prototype is one glove with a fixed vocabulary of 10 phrases. The language results come from one native-speaker reviewer per language.
- IV. The API key is kept in the browser only. Screenshots that expose the key should not be shared.
- V. Access to the N-ATLaS model is controlled by its publisher, so each reviewer needs an approved account.

A

Appendix: Vocabulary and Reference Phrases

The reference phrases are used as examples in the prompt and as the fallback when the server does not answer. N-ATLAS output can differ from them.

No.	English	Yorùbá	Hausa	Igbo
1	Good Morning	Ẹ kú àárọ̀	Ina kwana	Ụtụtụ ọma
2	My name is	Orukọ mi ni	Sunana	Aha m bụ
3	I am a student	Akẹkọọ ni mi	Ni ɗalibi ne	Abụ m nwa akwụkwọ
4	Hameed	Hameed	Hameed	Hameed
5	This is my work	Ìṣẹ mi ni eyi	Wannan shine aiki na	Nke a bụ ọrụ m
6	FUTminna	FUTminna	FUTminna	FUTminna
7	Please	Ẹ jọwọ	Don Allah	Biko
8	How are you	Bawo ni?	Yaya kake?	Kedu ka i mere?
9	I need help	Mo nilo iranlọwọ	Ina bukatar taimako	Achọrọ m enyemaka
10	Stop	Duro	Tsaya	Kwụsị